Fine-Grained Building Extraction With Multispectral Remote Sensing Imagery Using the Deep Model

Zhenqing Wang, Yi Zhou, Futao Wang, Shixin Wang, Gang Qin, Weijie Zou, Zhuochen Wang, Saimiao Liu, and Jinfeng Zhu

***Abstract*— Extracting buildings from remote sensing imagery can serve many industries. Single-category building extraction, however, has been unable to meet the actual needs of sectors, such as disaster prevention and mitigation. In this study, we create a large-scale high-quality multispectral dataset (BFE-Set) for fine-grained extraction of buildings. BFE-Set uses the GaoFen-2 multispectral satellite as the data source and divides buildings into three fine-grained categories according to their structure types: steel and reinforced concrete structure, masonry structure, and block stone structure. It includes 20 countries, where disasters often occur in China, and contains nearly 260 000 building instances. We propose the BFE-Net, a fully convolutional neural network that performs better on the BFE-Set for fine-grained extraction. The styles of buildings in different regions are quite different. Through the combined use of batch normalization (BN) and instance normalization (IN), BFE-Net enhances the model's style invariance. BFE-Net extracts feature from visible light and near-infrared data through a parallel feature extraction subnetwork and uses an attention mechanism to fuse the two modal features. This takes full advantage of the characteristic complementary effects of visible light and near infrared. In addition, knowledge distillation (KD) is used to fully mine the interclass relationships of fine-grained buildings, making the model more generalized in fine-grained classification. The experimental results show that BFE-Net is superior to other building extraction methods, and the fine-grained extraction effect is encouraging, which can provide a valuable reference. BFE-Set and BFE-Net will play a greater role in industries, such as disaster response and smart cities.**

***Index Terms*— Fine-grained building extraction, fully convolutional network, open multispectral dataset, remote sensing, semantic segmentation.**

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I. Introduction

BEING the principal locations of human production and living, buildings are an integral part of human beings. In many different sectors, including urban planning, real-estate management, disaster preparedness and mitigation, and others, the spatial distribution of buildings can be quite important [1], [2], [3]. Building footprints can be timely acquired using remote sensing satellites with excellent spatial resolution, which can immediately gather large-scale surface observation data.

Both supervised machine learning and unsupervised morphological building indices (MBIs) are commonly used in traditional building extraction techniques. By connecting implicit building features (such as brightness, size, and contrast) and morphological operators (such as top-hat transformation), the MBI extracts building footprints from images [4]. The accuracy of MBI is subpar due to the similarities between bare ground, roads, and other ground objects and buildings. Huang and Zhang [5] suggested the morphological shadow index (MSI) to constrain MBI in order to enhance its performance. To counteract the effects of narrow roads and bright greenery, employ a geometry index and a vegetation index. Unsupervised MBI means that it does not need to provide training samples or go through the laborious training phase. Unfortunately, the extraction accuracy is difficult to meet the requirements due to the subjective selection of the threshold and the interference of other ground objects. Building extraction frequently employs machine learning techniques, such as support vector machines, random forests, and artificial neural networks [6], [7], [8]. These works train and learn the features of a single pixel in the image. The extraction results are sloppy and there are a lot of incorrectly extracted pixels since the context information of the pixels is not taken into account. Several studies use machine learning and object-oriented approaches to address this issue [9], [10], [11]. Classify the superpixels that are made up of several nearby homogenous pixels and serve as the analysis object. The roughness of the extraction results is significantly reduced by using this technique to incorporate neighborhood features into machine learning. Also, the complexity of model reasoning is lowered by switching from pixels to superpixels as the analysis object. However, the precision of superpixels determines the extract effect, and they cannot analyze high-level semantic information.

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Deep learning has excelled recently in disciplines, such as natural language processing, image analysis, and video recommendation. It no longer just relies on the shallow features of the target but also on their strong feature extraction capabilities. The fully convolutional neural network used in the deep learning method has enhanced building extraction to a higher degree. Deep learning becomes increasingly and more dependent on training data as it advances to deeper architectures and more parameters. The model can learn more generic features if the training set contains enough data and there are enough different building representations. Four well-known large-scale open datasets are currently accessible for building extraction. These are the Massachusetts dataset [12], the ISPRS Vaihingen and Potsdam datasets (https://www2.isprs.org), the Inria dataset [13], and the WHU Building dataset [14]. The Massachusetts dataset includes 151 aerial photographs with a resolution of 1 m and a size of 1500 × 1500 pixels. However, it is of poor quality and has not been utilized in recent literature experiments. The Vaihingen and Potsdam datasets are the two subsets of aerial imagery datasets that make up the ISPRS dataset. There are 16 photograph slices of 1000–4000 × 1000–4000 pixels with a resolution of 0.09 m in the Vaihingen dataset and 38 image slices of 6000 × 6000 pixels with a resolution of 0.05 m in the Potsdam dataset. There are not enough building instances in the ISPRS dataset, which only encompasses a total area of 13 km^2. The Inria dataset, which spans 405 km^2 with a resolution of 0.3 m and contains buildings from five urban scenarios, can be utilized to evaluate the extrapolation and generalization skills of models to some extent. The high-quality WHU building dataset includes aerial and satellite imagery that spans areas of 450 and 550 km^2, respectively. Numerous building extraction works have been produced using the aforementioned available datasets. A SegNet variant was trained on remote sensing imagery of urban areas by Audebert et al. [15], who also added multikernel convolutional layers for quick prediction aggregation at multiple scales. A Siamese U-Net with shared weights in both branches was proposed by Ji et al. [14]. Also, using both the original and downscaled versions of the image as input greatly enhances segmentation accuracy. An edge-preserving convolutional neural network that integrates edge detection with the architecture extraction procedure was created by Guo et al. [16]. The extracted structures can keep their edge and positioning precision. A boundary refinement convolutional module was introduced by Guo et al. [17] to enhance building prediction by determining the angle from the center of each pixel in an image to the nearest object it might be a part of. In order to reach prior constraints, Wang et al. [18] combine edge features with additional learnable multiscale features at the feature level, then further improve the effect, and use the attention mechanism.

Due to the difference in building features between the training set and the test set in the aforementioned study, the model's actual generalization ability is, however, weak. Images from the extraction area and the training area are frequently cross-domain while extracting buildings in real life. The model must be capable of relatively high cross-domain extraction. Furthermore, more complex requirements in practical applications cannot be met by a single category of buildings. In seismic disaster events, casualties and property losses are primarily caused by building damage. Buildings of different structural types have varying abilities to withstand disasters, leading to varying degrees of destruction. Therefore, in earthquake risk management, not only the spatial distribution outline of buildings is needed but also other attribute information, such as building structures. Geiß et al. [19] aimed to achieve earthquake loss modeling by integrating multisensor remote sensing and machine learning to estimate building structural types. However, this method requires establishing different extraction rules for spectral and texture features based on various remote sensing image data, making it incapable of automatically extracting building structural information. Qi et al. [20] combined Google Earth imagery, Baidu Street View, crowdsourced data, and local building prior knowledge to infer building outlines and structures for addressing earthquake vulnerability tasks. The inclusion of various types of input data greatly limits the model's applicability, making it challenging to achieve satisfactory results when some input data are missing or difficult to obtain. Liu et al. [21] collected building structural category information through on-site surveys when conducting earthquake risk modeling. Although manual surveys ensure result credibility, for a large-scale research area, substantial human, material, and time resources are required. Besides, the majority of optical satellites can also collect near-infrared data on the ground; therefore, their importance as a supplement to visible light data cannot be understated.

In order to achieve this, we produced a sizable fine-grained extraction dataset (called BFE-Set); 20 building-concentrated locations where disasters frequently happen in China are covered by BFE-Set using Gaofen-2 multispectral images. Three categories are used to categorize buildings: 1) steel and reinforced concrete structure; 2) masonry structure; and 3) block stone structure. We use rigorous ground truth quality control and unified image preprocessing to assure the dataset's quality. For this data, we propose a building fine-grained extraction network (called BFE-Net). The encoder–decoder structure is used by BFE-Net. The encoder can increase style invariance and has feature extraction for visible and near-infrared data. The multimodal features that the encoder extracted are corrected and fused by the decoder. The knowledge distillation (KD) is used in the training strategy to thoroughly mine the interclass relations of fine-grained buildings.

II. Dataset

A. Image Collection and Preprocessing

We use the regionalization of natural disasters in China as the basis when choosing the image collection area in order to increase the dataset's value in the disaster field. Using the county as the fundamental unit, Lansheng et al. [22] carried out a systematic assessment of the severity of natural disasters based on the disaster system theory, regional differentiation theory, and geoscience information map theory. China's natural disaster zoning was developed based on this, and it consists of three levels: the first level, which consists of six natural disaster zones; the second level, which consists of 26 natural disaster areas, and the third level, which consists of 93 natural disaster areas. We chose the concentrated region

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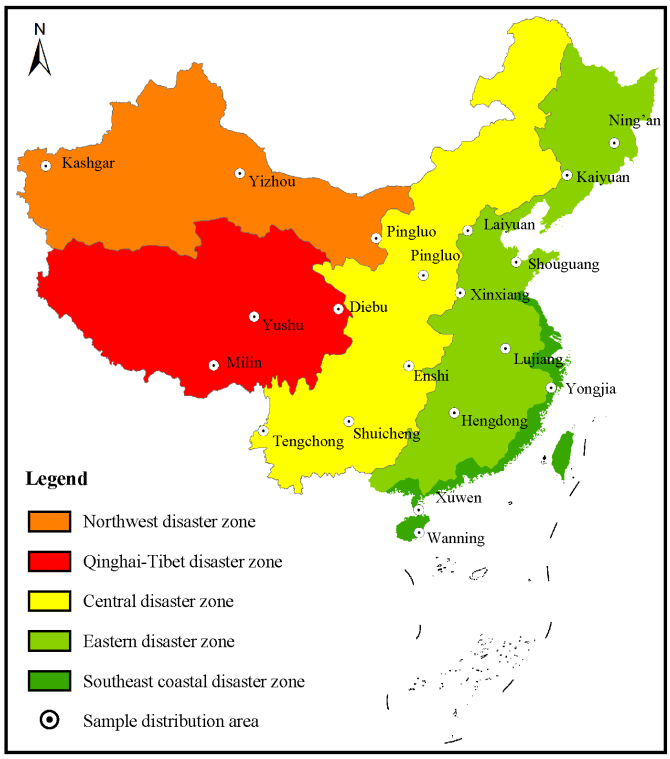


Fig. 1. Geographical distribution of images in BFE-Set in the comprehensive regionalization of natural disasters in China.

of buildings in 20 counties as the research area in BFE-Set based on the natural disaster zoning. In the five disaster zones on land, as depicted in Fig. 1, they are somewhat evenly dispersed. They include: Yongjia, Xuwen, and Wanning in the southeast coastal disaster zone; Ning'an, Kaiyuan, Laiyuan, Shouguang, Xinxiang, Lujiang, and Hengdong in the eastern disaster zone; Daning, Enshi, and Hengdong in the central disaster zone; Tengchong and Shuicheng, Kashgar, Yizhou, and Pingluo in the northwest disaster zone, and Diebu, Yushu, and Milin in the Qinghai-Tibet disaster zone.

Gaofen-2 data are the data source for BFE-Set. Gaofen-2 is the first civilian optical remote sensing satellite independently developed by China with a spatial resolution better than 1 m. It is equipped with two high-resolution 1-m panchromatic and 4-m multispectral sensors, and the subsatellite point spatial resolution can reach 0.8 m. High spatial resolution and a broad field of view, two features of the Gaofen-2 image, enable detailed ground information to be viewed across a wide area. Land cover surveys, disaster emergency assessments, and other significant applications have all made use of Gaofen-2.

We used several preprocessing techniques on the Gaofen-2 images to produce high-quality and high-resolution remote sensing imagery. Before choosing the dataset, we first evaluated the quality of the raw data and eliminated images with an excessive number of clouds, noise, or bright spots. Second, radiometric calibration and orthorectification are conducted on panchromatic images, whereas radiometric calibration, atmospheric correction, and orthorectification are performed on multispectral images. Lastly, we combine the panchromatic images with the multispectral resultant image using the Gram–Schmidt transform to increase the spatial resolution of the multispectral image. The final images have a spatial resolution of 0.8 m.

B. Fine-Grained Category Design

The technical specifications for the First National Natural Disaster Comprehensive Risk Survey issued by the Leading Group Office of the First National Natural Disaster Comprehensive Risk Survey of the State Council of China regulated the distinction of building structure types. Among them, the technical guideline for the survey of urban housing construction divides urban buildings into steel structure, reinforced concrete structure, masonry structure, wooden structure, and other structure [23]. The technical guidelines for the survey of rural housing construction divide the structural types of rural buildings into steel structure, reinforced concrete structure, block stone structure, civil structure, hybrid structure, cave dwellings, and other structure [24]. Considering the limited architectural features acquired by remote sensing images, it is unrealistic to identify all the above building categories from remote sensing images. Based on the similarities and differences in the features of different types of buildings, the buildings are divided into three different structural types: steel and reinforced concrete structure, masonry structure, block stone, and other structure. The fine-grained ground truth production standard of buildings is formed from the aspects of geometric features, spectral features, and spatial features of buildings through the combination of on-site ground survey and visual interpretation of Gaofen-2 images.

The Steel and Reinforced Concrete Structure: 1) geometric features: big single area, a high number of stories (approximately 20 stories), and irregular wall boundaries with apparent bumps; 2) spectral features: the inclined angle of view causes inconsistencies in color between the body and the top, and the shadow of the building is longer; and 3) spatial features: the monomers are distributed in dots, the distance between the monomers is relatively large, and the combination of multiple monomers is distributed in clusters; typically, parking lots, gardens, vegetation, and other auxiliary features exist.

The Masonry Structure: 1) geometric characteristics: the majority of them are buildings with fewer than six floors, regular limits, and sharp edges; 2) spectral characteristics: the building shadows are lengthy; and 3) geographical characteristics: the units are close together and multiple units form a large area of regular buildings in space.

The Block Stone and Other Structure: 1) geometric characteristics: the single body is relatively low and the area is tiny; 2) spectral characteristics: no shadow or shadow is not obvious; and 3) geographical characteristics: the distribution is scattered or dense, which is a relatively extreme distribution. Some will have courtyards, and the courtyards frequently have sundries in them.

The examples of buildings for the three fine-grained categories are shown in Fig. 2. After labeling, a total of 258 912 building instances were obtained, including 14 686 instances of steel and reinforced concrete structure, 51 446 instances of masonry structure, and 192 780 instances of block stone and other structure. We visualize the number and area distribution of each type of building, as shown in Fig. 3. The proportion of block stone structure buildings is significantly higher than that

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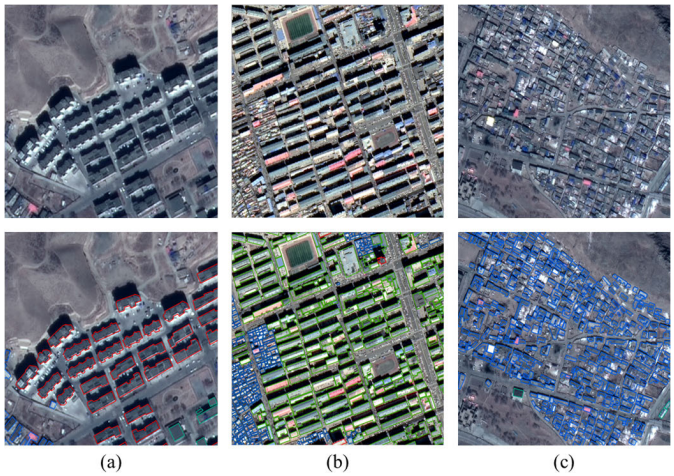


Fig. 2. Examples of buildings for fine-grained categories. Among them (a) steel and reinforced concrete structure, (b) masonry structure, and (c) block stone structure. The remote sensing imagery is shown in the first line, and the image with the building's polygonal annotation is shown in the second line.

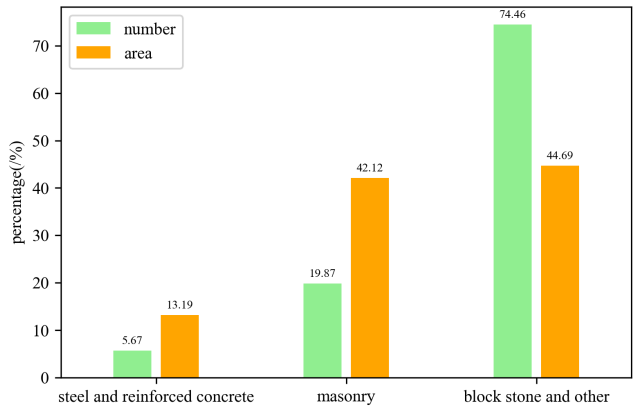


Fig. 3. Number and area distribution of buildings of different categories.

of reinforced concrete structure and brick-concrete structure. This long-tail phenomenon is in line with the real situation and brings challenges to fine-grained extraction tasks.

C. Ground Truth Quality Control

We used a quality control procedure to get extremely accurate ground truth. In addition to strictly following the criteria in Section II-B for interpretation, we have several additional rules in place. They include: for adjacent and well-defined buildings, there is a polygon annotation for each building; for adjacent buildings with blurred boundaries, multiple buildings can correspond to a polygon annotation; for the buildings where the body of the building is photographed, the roof is uniformly marked; obvious concave–convex boundaries and corners need to be marked; buildings above buildings do not need to be marked separately. The quality of the ground truth, including building boundaries and fine-grained categories, will then be further examined by data inspectors that we have invited to the project. The quality of remote sensing imagery and the accompanying ground truth will next be examined by professionals in the field.

D. Characteristics of the Dataset

The detailed parameter comparison between BFE-Set and four popular public datasets is shown in Table I, and its characteristics are summarized as follows.

Multispectral: The images in the BFE-Set additionally contain spectral information in the near-infrared region in addition to the visible light band. Only red, green, and blue bands are present in the Massachusetts, Inria, and WHU datasets and the ISPRS dataset only has four bands in some images. Near infrared response (NIR) data have been shown to be useful for building extraction in numerous studies. Subsequent experiments in this study will also prove that richer spectral information can lead to more accurate building extraction results.

Fine-Grained: As far as we are aware, the BFE-Set is the only large-scale open dataset that classifies buildings into multiple categories from a structural perspective. According to their structure, BFE-Set divides buildings into three categories. Buildings of various structure have significantly diverse levels of disaster resistance and usable time. The fine-grained design makes the task of extracting buildings from remote sensing play a greater role, particularly in the predisaster loss prediction and postdisaster loss assessment in the field of disasters.

Wide Distribution: Since it covers a total of 3396 km^2 in 20 key counties in various disaster zones in China, the wide distribution of BFE-Set is also a major feature. The wide area increases the diversity and regionality of the buildings. Buildings of different styles pose a huge challenge to the generalization of the model, which also makes the data have greater practical value.

Large Number: BFE-Set contains nearly 260 000 building instances, and the availability of large-scale labeled data provides data support for deep learning methods. Sufficient data also ensure high intraclass variance, including size variance, shape variance, color variance, angle variance, and background variance. In addition, each region has at least two time-phase images, and different seasons and weather conditions will bring different imaging differences.

E. Dataset Splits

BFE-Set is divided into a training set, a validation set, and a test set. Images from various cities are used in each set. Wanning, Ning'an, Laiyuan, Xinxiang, Hengdong, Daning, Enshi, Tengchong, Kashgar, Pingluo, and Yushu are among the cities in the training set. The validation set includes the following cities: Yongjia, Lujiang, Diebu, and Yizhou. The test set consists of the following cities: Xuwen, Milin, Kaiyuan, Shuicheng, and Shouguang. We had to crop large-scale remote sensing imagery because of memory limitations. Each slice is 512 $\times$ 512 in size, and adjacent slices repeat 20%; 7579 training slices, 1572 validation slices, and 1334 test slices were ultimately collected.

III. Methodology

Buildings have different scales. In order to make the model have multiscale feature fusion function, we use HR-Net [25] as the baseline. Adding IBN-block improves the style invariance of the model. Visible and near-infrared data are extracted using

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TABLE I
COMPARISON OF PARAMETERS BETWEEN BFE-SET AND OTHER DATASETS

Dataset name	GCD / (m)	Area / (km^2)	Spectrum	Number of categories	Dataset size / (number * row * column)
Massachusetts	1.0	340	R-G-B	1	151*1,500*1,500
ISPRS	0.05/0.09	2/11	IR-R-G-B/ IR-R-G	1	16 * 1,000~4,000 * 1,000~4,000/ 38 * 6,000 * 6,000
Inria	0.3	405	R-G-B	1	180 * 5,000 * 5,000
WHU	0.075/0.3-2.5	450/550	R-G-B	1	8,188 * 515 * 512
BFE-Set	0.8	3396	IR-R-G-B	3	10,485 * 515 * 512

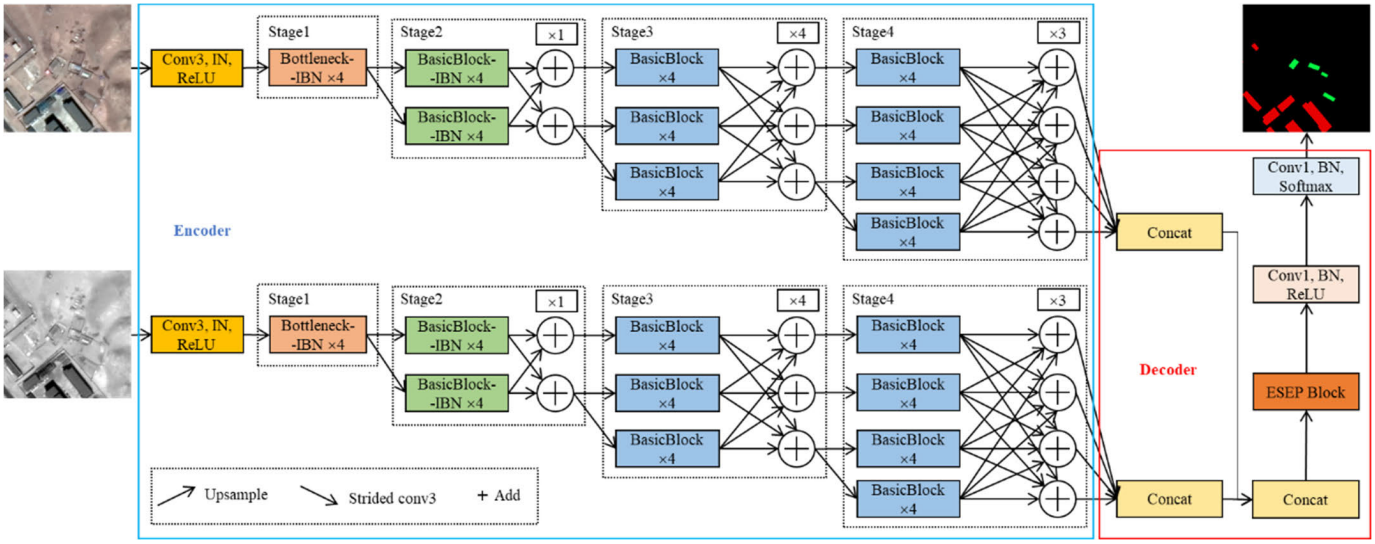


Fig. 4. Structure of BFE-Net. The structure in the blue box is the encoder, and the structure in the red box is the decoder.

an encoder consisting of two feature extraction networks that do not share weights. Afterward, the features of the extracted two modal data are fused and corrected by the decoder, and the extraction results are output through the Softmax classifier. Finally, KD is used to mine the interclass associations of fine-grained buildings to further optimize the fine-grained extraction results. This network that can extract fine-grained buildings is named BFE-Net, and its structure is shown in Fig. 4.

A. Multimodal Feature Extraction Encoder With Style Invariance

Due to the differences of different buildings and the influence of factors, such as light intensity, the styles of buildings in images in different regions will be different. The so-called style includes shallow features, such as color, brightness, and texture. We need to make the model not cause large output differences due to differences in building styles. In order to increase the style invariance of the network, a change has been made in the normalization of features. In addition to the commonly used batch normalization (BN), we additionally add an instance normalization (IN) to the feature extraction network.

BN [26] is to calculate the mean and standard deviation of all samples in a batch, each channel is calculated independently, and the output statistics will be affected by all samples in the batch. BN preserves the distinction between individual samples but also makes the model vulnerable to appearance changes. Unlike BN, IN [27] computes the mean and standard deviation for each channel of each sample

$$\begin{cases} \hat{x}_{nchw} = \dfrac{x_{nchw} - \mu_{nc}}{\sqrt{\sigma_{nc}^2 + \varepsilon}} \\ \mu_{nc} = \dfrac{1}{HW} \displaystyle\sum_{h=1}^{H} \sum_{w=1}^{W} x_{nchw} \\ \sigma_{nc}^2 = \dfrac{1}{HW} \displaystyle\sum_{h=1}^{H} \sum_{w=1}^{W} (x_{nchw} - \mu_{nc})^2 \end{cases} \tag{1}$$

where x_{nchw} represents the values of the cth channel, hth row, and wth column of the nth sample of the input data. H and W are the width and height.

IN eliminates individual contrasts, while reducing useful global information and affecting classifier feature recognition. The general conclusion is that BN is suitable for cognitive tasks (such as image classification, semantic segmentation, object detection, etc.) and IN is suitable for generation tasks

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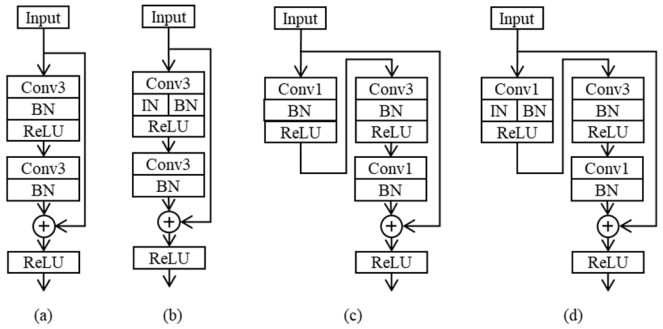


Fig. 5. Structure of network basic modules. BasicBlock and Bottleneck are the basic modules in the original HR-Net, and BasicBlock-IBN and Bottleneck-IBN are the basic modules of the proposed method. (a) BasicBlock. (b) BasicBlock-IBN. (c) Bottleneck. (d) Bottleneck-IBN.

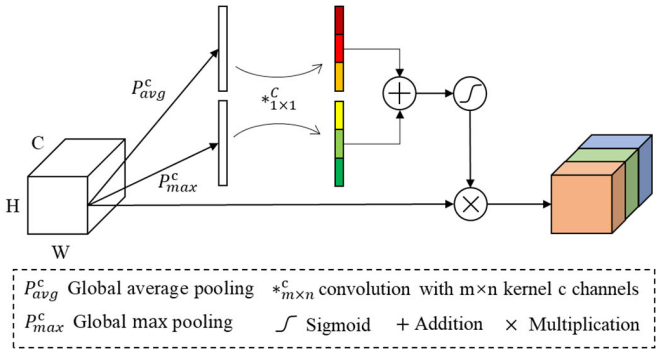


Fig. 6. Structure of ESEP module.

(style transfer, image denoising, and image generation). When the BN of the model is replaced by IN in cognitive tasks, it is usually found that the performance drops significantly, but this is not because IN is not beneficial to cognitive tasks, but because it is not used correctly. In fact, IN brings the invariance of the feature mean and variance statistics between different samples, while BN, as a global normalization method, retains the differences between different samples. The invariance brought by IN is what we want: we hope that the magnitude of changes in the appearance factors, such as brightness, color, and style of the input image, will not affect the prediction results of the model, that is, style invariance, but it is also what we do not want: since the model's ability to correctly classify objects depends on its ability to distinguish between various object types based on their depth features. This distinction is largely reflected in the mean and variance. We do not want to use IN to destroy this distinction. Therefore, we use IN for some features in the first half of the model, and the deep layer still keeps BN unchanged, so that the style invariance is designed as a priori in the model.

Inspired by IBN-Net [28], in order to reduce the feature variance caused by the appearance of shallow layers, while not interfering with the content recognition of deep layers, we only add IN in the shallower parts of the model. As shown in Fig. 4, the Bottleneck in stage 1 and the BasicBlock in stage 2 in the original HR-Net are replaced by Bottleneck-IBN and BasicBlock-IBN, respectively. The structure of Bottleneck-IBN and BasicBlock-IBN is shown in Fig. 5. While retaining BN (to ensure that content-related information can be passed into the deep layer smoothly), adding IN improves the robustness to style changes.

For optical data and near-infrared data, we employ parallel independent feature extraction. It is inappropriate to directly input the data of the two modalities into a backbone by stacking channels. On the one hand, it may interfere with the feature extraction of optical data (the ImageNet pretraining model is trained based on red-green-blue (RGB) data); on the other hand, it may weaken the auxiliary role of near-infrared data. Therefore, we use two backbones with the same structure but not shared weights for feature extraction and fully feature mining the data of the two modalities.

B. Multimodal Feature Fusion Decoder

We use decoder to fuse the multimodal features extracted by encoder. The feature groups of the optical data and the feature groups of the near-infrared data are concatenated in the channel dimension, and then, the convolutional layers are used to fuse these features and then use the attention module named effective squeeze-excitation plus (ESEP) to modify the weight of each feature to further improve the feature representation ability. Finally, 1×1 convolution is used to reduce the feature dimension, and the final fusion feature result is obtained, which is used for classification by Softmax.

ESEP is channel attention that can improve squeeze-excitation (SE) more effectively, and its structure is shown in Fig. 6. SE is a commonly used channel attention method to explicitly model the interdependence between feature map channels. SE compresses the spatial correlation by global average pooling to learn channel-specific descriptors and then uses two fully connected (FC) layers along with the Sigmoid function to rescale the input feature map to highlight useful channels. However, SE modules have certain limitations. To avoid the high model complexity burden, the two FC layers of the SE module need to reduce the channel size, and this channel dimensionality reduction will lead to loss of channel information. Therefore, ESEP uses only one FC layer to preserve channel information to the greatest extent and improve performance. In addition, SE uses global average pooling when compressing spatial correlation. Although it can take global information into account, it weakens locally important saliency information. Therefore, in addition to the global average pooling, we use the global max pooling. After the two kinds of pooling, the weights of the FC layers are shared, so that the model can improve the learning ability of the model while ensuring that the complexity is almost unchanged.

C. KD for Mining Relationships Between Classes

In the field of deep learning, data are usually marked as hard labels, but in fact, a piece of data contains different types of information, directly labeling it as a hard label will lead to the loss of a lot of information, which will affect the effect of the model. There is a correlation between buildings of different categories, and the hard label will weaken the interclass correlation, and there is a risk of overfitting. Label smoothing [29] can generate soft labels, but soft labels are artificially set and have strong subjectivity. The main difference between KD [30] and label smoothing is that the soft label of KD is obtained through network inferencing. The teacher model in KD counts the effective information of the dataset, retains

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	probability^{s}	probability^{m}	probability^{b}	probability^{o}
hard label	1	0	0	0
soft label	0.89	0.1	0.01	0

Fig. 7. Difference between hard label and soft label. Probability^{s} represents the probability that the pixel belongs to the steel and reinforced concrete structure. Probability^{m} represents the probability that the pixel belongs to the masonry structure. Probability^{b} represents the probability that the pixel belongs to the block stone structure. Probability^{o} represents the probability that the pixel belongs to other objects.

the association information between classes, and eliminates some invalid redundant information. Compared with label smoothing, the soft label trained by the model on the dataset is more reliable.

The conventional practice in training models involves aligning the model's softmax distribution with the ground truth (hard labels). However, the KD method matches the softmax distribution of the student model with the class probabilities output by the teacher model (soft labels). The latter approach holds certain advantages over the former: the softmax distribution of a well-trained teacher model contains a certain degree of knowledge. The outputs of the teacher model's softmax layer, apart from positive instances, also carry substantial information about negative labels. For instance, the probabilities associated with certain negative labels are significantly higher than others. In contrast, in traditional training processes, all negative labels are treated uniformly (with probabilities as 0). This implies that the training method of KD imparts a greater amount of interclass relationship information to the student model from each sample compared to the conventional training method.

As depicted in Fig. 7, in the context of fine-grained building extraction tasks, hard labels can only inform us that the red pixels in the image belong to steel and reinforced concrete structures, rather than masonry or block stone structures. A trained teacher model might indicate that a pixel is most likely associated with steel and reinforced concrete structures, with a lower probability of it being related to masonry structures, and it is highly unlikely to be associated with block stone structures. Once this probability relationship among different categories is transferred from the teacher model to the student model, the student model exhibits enhanced generalization in fine-grained category recognition.

The teacher model in this work is BFE-Net with HRNet48 as the baseline, and the student model is BFE-Net with HRNet18 as the baseline. We first train the teacher model, and the loss function is the cross entropy of the model result

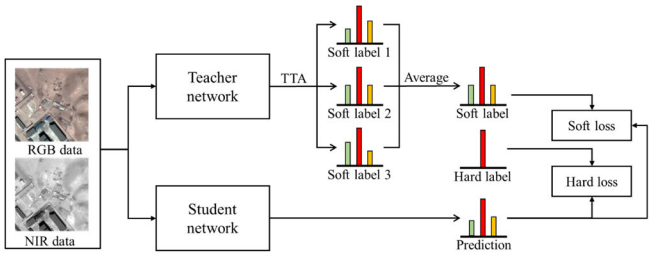


Fig. 8. Flowchart of KD.

and the hard label. As shown in Fig. 8, the loss function of the student model contains two loss branches, one is the distillation loss L_{soft}, and the other is the regular loss L_{hard}. When soft labels are obtained by the teacher model, test-time augmentation (TTA) is used to increase the reliability of the soft labels. The TTA includes horizontal flip, vertical flip, and 90° rotation. The distillation loss enables the student model to acquire rich interclass knowledge embedded within the teacher model. However, the teacher model is not devoid of errors, and if the training relies solely on distillation loss, these errors might propagate to the student model. The concurrent usage of regular loss functions mitigates the adverse impact of certain erroneous knowledge from the teacher model, thereby ensuring a more comprehensive learning process for the student model. The loss function of the student model is as follows:

$$\begin{cases} L = \alpha L_{\text{soft}} + \beta L_{\text{hard}} \\ L_{\text{soft}} = -\dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\sum_{c=1}^{M} s_c^i \log\left(p_c^i\right) \\ L_{\text{hard}} = -\dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\sum_{c=1}^{M} y_c^i \log\left(p_c^i\right) \end{cases} \tag{2}$$

where N is the number of pixels when calculating the loss, M is the number of categories, and s_c^i is the probability (soft label) that the ith pixel output by the teacher model belongs to category c. p_c^i is the probability that the ith pixel output by the student model belongs to category c. y_c^i is the probability (hard label) that the ith pixel in the ground truth belongs to category c.

IV. Experimental Settings and Results

A. Experimental Parameter Settings

All experiments in this study were trained on Nvidia GeForce RTX 3090 GPU, and models were built using Pytorch [31]. Both BFE-Net and the backbone in the comparison method use the pretrained model trained by ImageNet [32]. The total number of training epochs for the experiment is 100, and the batch size is 8. The initial learning rate is 0.001. The optimizer uses Adam with weight decay (decay factor is 0.001). Cosine annealing is used to adjust the learning rate, the minimum learning rate is set to 0.00001, and five rounds of warmup are used to ensure the stability of the training. The random seed in the experiment was set to 42.

B. Evaluation Metrics

The F_1 score is used to evaluate the performance of the model for extracting each type of building and background.

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TABLE II
ACCURACY RESULTS OF BFE-NET AND OTHER MODELS

Model	F_1-0 / (%)	F_1-1 / (%)	F_1-2 / (%)	F_1-3 / (%)	OA / (%)
ResU-Net	95.69	47.78	70.45	35.65	91.11
DeeplabV3+	95.64	43.91	63.91	34.98	90.80
MEC-Net	95.94	49.49	72.39	45.64	91.74
BFE-Net	96.48	55.73	75.71	55.42	92.47

The calculation formula for F_1, which is the harmonic mean of recall and precision, is as follows:

$$F_1 = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \tag{3}$$

where precision is the ratio of true positive to the number predicted to be positive, and recall is the ratio of true positive to the number of actual positive. The overall accuracy (OA) is used to evaluate the overall performance of the model. OA is the ratio of the number of correctly classified pixels to the total number of pixels in the sample.

C. Comparison Methods

The following three state-of-the-art building extraction methods are selected for comparison with our method. ResU-Net [33]: combining the advantages of residual learning and U-Net, using fewer parameters to achieve better performance. DeeplabV3+ [34]: using a spatial pyramid pooling module with dilated convolutions to introduce multiscale information and improve recognition performance. MEC-Net [18]: efficient fusion of prior edge features and learnable multiscale features.

D. Experimental Results

We perform a quantitative evaluation of BFE-Net as well as the comparative methods, and the evaluation results are shown in Table II. F_1-0 represents F_1 of the background (area other than the building). F_1-1 represents F_1 of steel and reinforced concrete structure buildings, F_1-2 represents F_1 of masonry structure buildings, and F_1-3 represents F_1 of block stone structure buildings. BFE-Net achieves the highest F_1 on all three categories of building extraction. OA is 1.36%, 1.67%, and 0.73% higher than ResU-Net, DeeplabV3+, and MEC-Net, respectively. Steel and reinforced concrete structure have the highest F_1, which may be related to the most obvious features of this category of building. The shadow of masonry structure buildings is also relatively obvious, which is easy to be confused with steel and reinforced concrete structure buildings, resulting in a low F_1. Some block stone structure buildings are similar to other surrounding features, so it is difficult to extract them effectively, so their F_1 is the lowest.

In order to show the effect of the model more intuitively, we visualize the inference results of some test sets, as shown in Fig. 9. In the first row, ResU-Net, DeeplabV3+, and MEC-Net recognize some steel and reinforced concrete structure as masonry structure. In the second row, there are also some misidentifications. This shows that BFE-Net's ability to divide different types of buildings is better than other models. In the third and fourth rows, ResU-Net and DeeplabV3+ are significantly inferior to MEC-Net and BFE-Net for the edge area of the building. In the fifth row, BFE-Net can still achieve better extraction results for block stone structure with poor imaging quality.

V. ANALYSIS

A. Analysis of BFE-Set

To investigate the causes of fine-grained building misidentification, we computed the precision matrix and recall matrix for the model's inference results, as illustrated in Fig. 10. It can be observed that for steel and reinforced concrete structure buildings, the proportion of masonry structure is the largest in the pixels that are misidentified as this category, and the proportion of masonry structure is also the largest in the pixels that are misidentified as other types of buildings. Some reinforced concrete and masonry structures are visually similar, thereby increasing the complexity of classification. Exploring effective feature enhancement methods and incorporating multimodal data (such as SAR and stereo mapping data) become imperative for improving classification performance. Masonry structures exhibited relatively high precision and recall, particularly in terms of recall. The misrecognized area of this type of building is mainly the pixels of the boundary area of the building. The correct classification of pixels in object boundary regions has always been a challenge. The block stone buildings are mainly easily confused with the background. Block stone buildings are smaller in size and occupy fewer pixels, resulting in less distinct features, and block stone buildings are mostly distributed in rural villages, which are easily affected by sundries. Detecting brick structure buildings in satellite images remains challenging. Future plans involve supplementing the dataset with higher resolution aerial remote sensing images to enhance the accuracy of block stone building detection.

The datasets of Massachusetts, ISPRS, Inria, and WHU all categorize buildings into a single class. To maintain consistency, we unified the three fine-grained categories in the BFE-Set into a single class for quality comparison with these datasets. Additionally, the testing set of BFE-Set is cross-domain, while the testing sets of the other four datasets have a style similar to their training sets. For a fair comparison, apart from the data split strategy introduced in Section II, an additional data split was performed. The data from each county were divided into training, validation, and testing sets in a 6:2:2 ratio and then merged, resulting in independent yet stylistically similar testing, validation, and training sets. The additional split data are denoted as BFE-Set-A. Uniform training and evaluation were conducted using ResU-Net to obtain the F_1 scores corresponding to different datasets as presented in Table III.

The F_1 score of building extraction in the BFE-Set dataset reached 82.69%, which is the lowest accuracy among all the datasets. However, it is important to note that the BFE-Set's test set is cross-domain, while the test sets of other datasets are relatively similar to their training sets. Hence, it is reasonable for other datasets to achieve higher F_1 scores than

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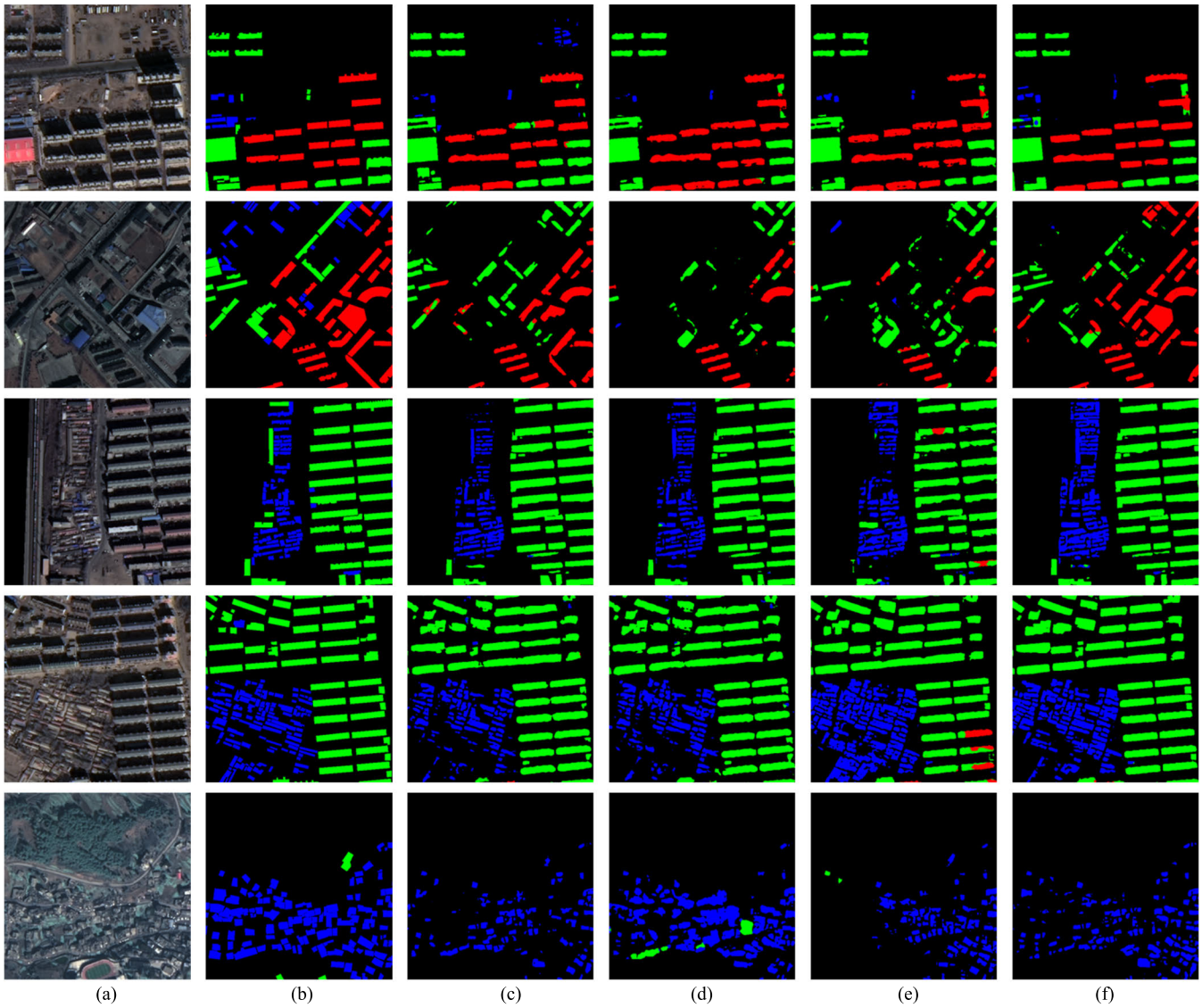


Fig. 9. Inference results of BFE-Net. Red pixels represent steel and reinforced concrete structure, green pixels represent masonry structure, and blue pixels represent block stone and other structure. (a) RGB data. (b) Ground truth. (c) Deeplabv3+. (d) ResU-Net. (e) MEC-Net. (f) BEF-Net.

TABLE III
F_1 CORRESPONDING TO DIFFERENT DATASETS

Dataset	Massachusetts	ISPRS	Inria
F_1 / (%)	84.13	95.50/96.52	88.42
Dataset	WHU	BFE-Set	BFE-Set-A
F_1 / (%)	94.43	82.69	88.51

the BFE-Set due to the domain shift. For BFE-Set-A, which follows similar dataset split rules as other datasets, the F_1 score for building extraction reached 88.51%. With the modified data split, the F_1 score surpasses that of the Massachusetts and Inria datasets. The F_1 scores for both the ISPRS and WHU datasets exceed 94%, but these datasets contain less complex building types, making it challenging to evaluate the model's true generalization ability. Notably, even in the case of BFE-Set, which encompasses complex building types and a cross-domain test set, it still achieves commendable performance, highlighting the high quality of the dataset.

B. Analysis of BFE-Net

To ascertain the effectiveness of BFE-Net, a series of ablation experiments were conducted, as presented in Table IV. These experiments were based on the HR-Net baseline, progressively integrating components, such as the IBN block, NIR processing module, and attention mechanism. The objective was to investigate the individual contributions of these components within the model.

The addition of IBN blocks greatly improves the extraction accuracy of the three types of buildings (4.32%, 1.73%, and 6.82%, respectively). The IBN blocks improve the style invariance of the model through a special treatment of feature normalization, enabling the model to achieve more generalized results on cross-domain data. We conducted experiments with and without IBN on the BFE-Set-A dataset to test whether

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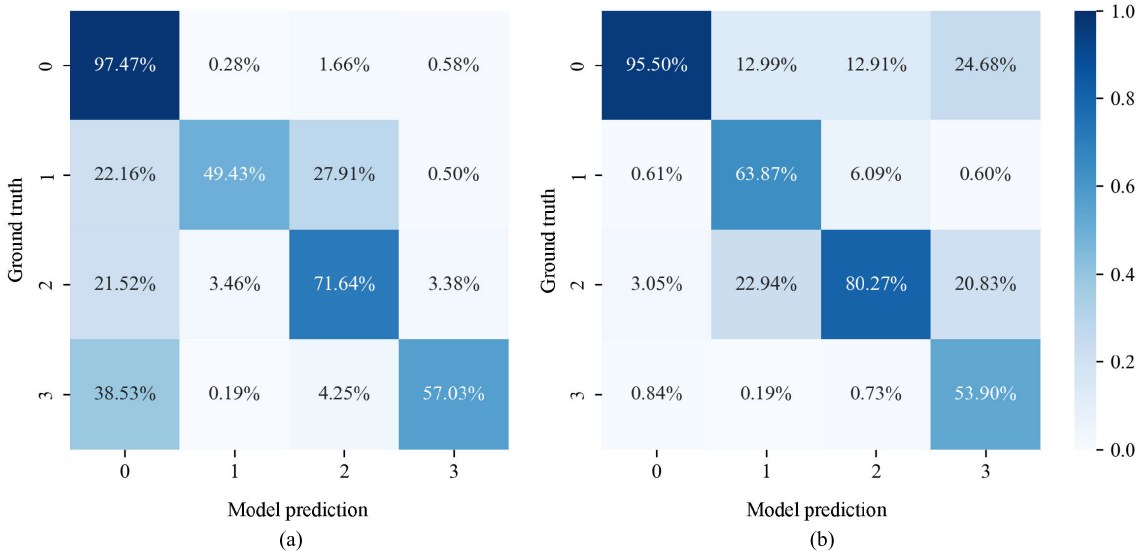


Fig. 10. (a) Precision matrix and (b) recall matrix of the model's inference results.

TABLE IV
ABLATION EXPERIMENTAL RESULTS

HR-Net	IBN	NIR	Attention	KD	F_1-0 / (%)	F_1-1 / (%)	F_1-2 / (%)	F_1-3 / (%)	OA / (%)
√	-	-	-	-	95.58	46.24	67.43	36.58	90.73
√	√	-	-	-	95.89	50.56	69.16	43.40	91.36
√	√	data fusion	-	-	95.89	51.45	69.22	45.78	91.47
√	√	feature fusion	-	-	95.90	52.85	70.47	50.07	91.61
√	√	feature fusion	SE	-	96.24	52.20	73.07	54.00	92.06
√	√	feature fusion	ESEP	-	96.37	54.36	73.53	53.99	92.29
√	√	feature fusion	ESEP	√	96.48	55.73	75.71	55.42	92.47

IBN can still maintain its advanced nature when dealing with data similar in style to the training data. HR-Net achieves 87.21% of the F_1 scores. HR-Net with IBN added achieves an F_1 score of 88.79% with an increment of 1.58%. A style-invariant model can enhance its robustness and generalization ability by allowing the model to ignore style changes in the data and focus more on learning the content features of the data. This means that if the test set and training set are of the same style, the model can still accurately identify and understand important information in the data.

In the processing of NIR data, the OA of using the input channel stacking method (data-level fusion) is 0.11% higher than that of not using NIR data. This shows that the feature supplement of NIR data improves the effect of fine-grained extraction of buildings. While using BFE-Net (feature-level fusion) for NIR processing, the OA is improved by 0.14% compared with channel stacking, which shows that using two feature extraction branches to process visible and NIR data separately is effective.

The use of SE improves OA by 0.45%, demonstrating the effectiveness of using attention in the decoder to correct multimodal features. ESEP improved the effect even more, increasing OA by 0.68%. The parameter quantity of SE is 0.65 M, and the parameter quantity of ESEP is 1.3 M. A small amount of parameter increase brings a certain effect improvement. ESEP makes the learned features more reasonable by preserving channel information to the greatest extent and adding local saliency information.

The use of KD increases the F_1-1, F_1-1, and F_1-3 of BFE-Net by 1.37%, 2.18%, and 1.43%, respectively. We apply KD to the comparison models to verify the generality of KD in fine-grained building extraction, and the results are shown in Table V. KD has produced different degrees of gain effects on the comparison models. We visualize the soft labels of the training set, as shown in Fig. 11. Compared with hard labels, soft labels in KD contain rich interclass knowledge. Through the transfer of knowledge between classes, the model can better distinguish fine-grained categories.

We conducted an analysis of different models in terms of floating point operations (FLOPs), params, training time, and inference time, as shown in Table VI. BFE-Net exhibits relatively lower FLOPs and parameter counts, indicating a comparatively smaller storage footprint. This aspect proves advantageous in deployment scenarios, where storage

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TABLE V
RESULTS OF APPLYING KD TO THE COMPARISON MODELS

model	KD	F_1-0 / (%)	F_1-1 / (%)	F_1-2 / (%)	F_1-3 / (%)	OA / (%)
ResU-Net	-	95.69	47.78	70.45	35.65	91.11
ResU-Net	√	95.93	50.51	72.45	47.78	91.77
DeeplabV3+	-	95.64	43.91	63.91	34.98	90.80
DeeplabV3+	√	95.80	51.26	67.69	39.73	91.14
MEC-Net	-	95.94	49.49	72.39	45.64	91.74
MEC-Net	√	96.09	53.14	73.15	45.29	91.93

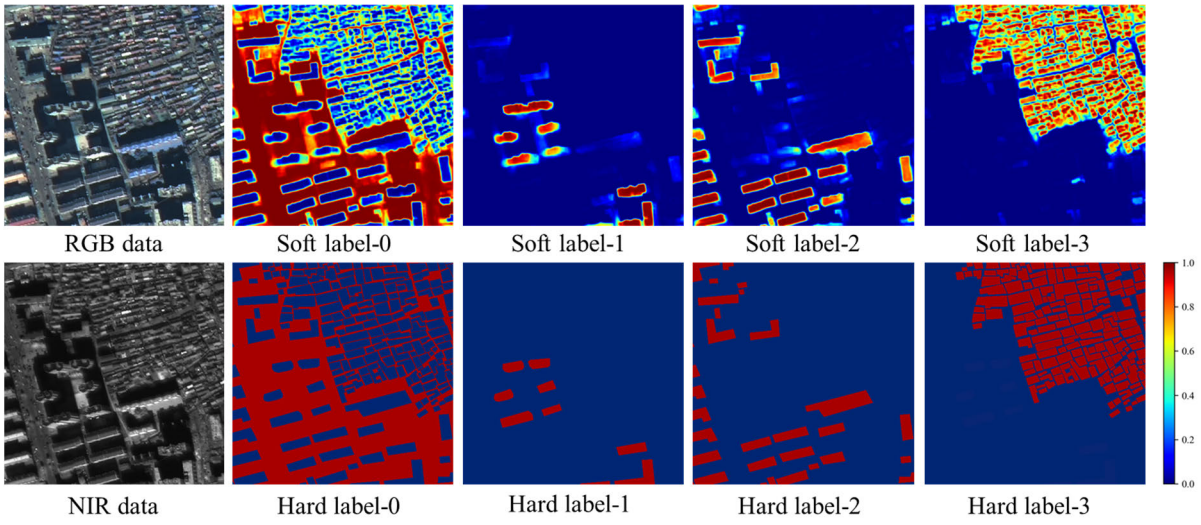


Fig. 11. Soft and hard labels for the training set.

TABLE VI
DIFFERENT MODELS IN TERMS OF FLOPS, PARAMS, TRAINING TIME, AND INFERENCE TIME

model	FLOPs / GMac	Params / M	training time / h	inference time / m
ResU-Net	42.96	32.52	7.79	1.02
DeeplabV3+	36.99	26.68	10.61	1.08
MEC-Net	62.11	30.71	24.25	1.75
BFE-Net	35.88	19.49	34.89 + 46.69(teacher)	2.08

resources are constrained. However, due to the incorporation of two encoders and the use of KD, the model's training time is notably extended in comparison to other models. It is noteworthy that KD exclusively affects the training phase, while practical application places more emphasis on inference efficiency. Given that BFE-Net involves substantial feature concatenation operations within its two encoders, the model's inference time is approximately twice that of other models. Leveraging the presence of multiple GPUs, if available, can potentially accelerate the inference process by assigning each encoder to a separate GPU.

Influenced by different cultures and histories, buildings in various cities possess unique characteristics. In addition to the IBN block employed in our method, which alleviates to some extent the impact of cross-domain data, the introduction of discriminators in adversarial learning [35] and the alternating optimization with entropy-minimization strategy [36] has been demonstrated to yield benefits in heterogeneous area. Subsequent efforts can be directed toward implementing similar measures on the BFE-Net architecture, thus enabling it to achieve improved performance when transferred to large-area and heterogenous area.

VI. CONCLUSION

In this work, we provide BFE-Set, a multispectral remote sensing fine-grained buildings dataset. The characteristics of multispectral, fine-grained, wide distribution, and large quantity make it have greater practical significance. The results of

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fine-grained building distribution will make the postdisaster relief needs clearer, manpower deployment will be more reasonable, and more lives will be saved. We also design a fully convolutional network BFE-Net for fine-grained building extraction. BFE-Net has the following advantages.

1) The joint use of BN and IN improves the style invariance of the model.
2) Perform parallel feature extraction on visible light and near-infrared modal data, and use ESEP to fuse multimodal features, fully exploiting the feature supplement capability of near-infrared data.
3) KD excavates the potential interclass relationship of fine-grained buildings, making the model more generalizable in fine-grained classification.

Experimental results obtained using BFE-Set demonstrate that BFE-Net is effective in building fine-grained extraction capabilities on multispectral datasets. In the future, we will consider further expanding the data volume of BFE-Set and adding other modal remote sensing data, such as SAR or stereo mapping data. For the convenience of researchers, BFE-Set will be available online at https://github.com/WangZhenqing-RS/BFE-Set.

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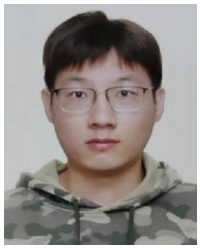
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